Full Interaction Log: EFF 2017 Analysis Task

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Model: Claude Opus 4 (Anthropic), via Claude Code CLI

Session type: Multi-turn interactive session

Prompt 1

User:

> read claude.md file

Model actions:

- Read `/Users/sid/Desktop/Git/cunef rojo/CLAUDE.md`
- Returned a summary of the project context: predoctoral RA at a policy school, R as primary language, EFF project in analysis + writing phase, preferences for commented code and explaining errors before fixes.

Prompt 2

User:

> Create these directories: code/, output/figures/, output/tables/, log/ Write 00_master.R that loads haven, tidyverse, fixest, modelsummary, boot, marginaleffects, sets path variables for data_dir/fig_dir/tab_dir, reads eff.dta and secondlang_prob.dta, then sources scripts 01 through 06.

Model actions:

1. Read the R coding style guide at `/Users/sid/Desktop/Main/r_coding_style_prompt.md` (read in three chunks: lines 1-80, 80-160, 160-240, 240-340).
2. Listed `toshare/` directory to confirm data file locations.
3. Created four directories via `mkdir -p`:
    - `code/`
    - `output/figures/`
    - `output/tables/`
    - `log/`

4. Verified directory tree with `ls -R`.
5. Wrote `code/00_master.R` with:
    - Six library() calls in usage order (not alphabetical, per style guide).
    - Path variables: root, data_dir (pointing to toshare/), fig_dir, tab_dir, log_dir.
    - read_dta() for both .dta files.
    - Six source() calls for scripts 01-06.
    - No cat()/message() calls, no section decorators (per style guide).

Files created: `code/00_master.R`

Prompt 3

User:

> Write 01_imputations_weights.R. I have already worked out the logic:
>
> - Q1a: w_adj = facine3/5, weighted.mean(totnet, w_adj)
> - Q1b: show sum(facine3) vs sum(w_adj) and total wealth implied by each -- this demonstrates the 5x inflation in population total, not in the mean itself
> - Q1c: weighted mean of totnet by imp using original facine3 (one implicate at a time)
> - Q1d: Rubin's rules -- Q_bar = mean of the 5 estimates from Q1c. B = var() of those 5 estimates. W = mean within-implicate variance computed as weighted variance of totnet divided by effective sample size. T = W + (1 + 1/5)*B. SE = sqrt(T). Save two CSVs: one for Q1b comparison, one for Q1d result.

Model actions:

1. Inspected eff.dta via Rscript:
    - 32,065 rows, 56 columns
    - 5 implicates (imp 1-5), 6,413 unique households
    - facine3 summary: min 2.19, max 53,337
    - totnet summary: min -735,359, max 631,363,007
2. Wrote `code/01_imputations_weights.R` implementing all four sub-questions.
3. Ran the script via Rscript to verify:
    - Pooled mean: 251,897 EUR (same under both weight schemes)
    - Population total inflated 5x: 92.7M vs 18.5M
    - Rubin's SE: 32,394
    - B (955,202) negligible relative to W (1.05 billion)
4. Confirmed two CSVs saved:
    - `output/tables/q1b_weight_comparison.csv`
    - `output/tables/q1d_rubins_rules.csv`

Files created: `code/01_imputations_weights.R`, 2 CSVs

Prompt 4

User:

> Write 02_descriptive_stats.R. Design decisions already made:
>
> - Age bins: cut(age_resp, breaks=seq(25,85,5), right=FALSE, labels=paste0(seq(25,80,5),"-",seq(29,84,5))). Filter age_resp 25-80.
> - Q2a: weighted homeownership rate (own) by age_grp using facine3
> - Q2b: ggplot with 3 series (All, Men where gender_resp==0, Women where gender_resp==1), theme_bw, scale_color_manual with black/blue/red, save PDF width=7 height=4.5
> - Q2c: boot package, R=500, set.seed(42), weighted median via cumulative weight, boot.ci type="perc", ribbon plot saved as PDF
> - Q2d: function plot_ownership_rate(data, outcome_var, xlab, ylab, title) using .data[[outcome_var]] syntax. Demonstrate on own_ot.

Model actions:

1. Checked variable coding via Rscript:
    - age_resp range: 19-86
    - gender_resp: 0 (man), 1 (woman)
    - own/own_ot: binary 0/1
    - Confirmed variable labels from Stata metadata
2. Checked filtering: 30,440 obs with age 25-84
3. Wrote `code/02_descriptive_stats.R` with all four sub-questions.
4. Ran end-to-end verification:
    - Homeownership rises from 27.5% (25-29) to 84.7% (65-69)
    - Bootstrap medians trace hump shape peaking at 211,584 EUR (60-64)
    - Three PDFs saved successfully
5. Confirmed outputs:
    - `output/figures/q2b_homeownership_age_gender.pdf`
    - `output/figures/q2c_median_wealth_bootstrap.pdf`
    - `output/figures/q2d_other_realestate_ownership.pdf`

Files created: `code/02_descriptive_stats.R`, 3 PDFs

Prompt 5

User:

> Write 03_homeownership_models.R. All decisions are made:
>
> - Mutate educ_cat: low if educ_resp<=4, medium if <=8, high if %in% c(9,11,12,1001,1002), else NA. Factor with levels low/medium/high.
> - log_hhinc = log(pmax(hh_inc, 1))
> - Q3a: feols(own ~ age_resp + I(age_resp^2) + educ_cat + emp_resp + self_resp + une_resp + hhsize + log_hhinc, weights=~facine3, vcov="hetero")
> - Q3b: same formula with glm(..., family=binomial("probit"), weights=facine3). avg_slopes() saved to CSV. modelsummary gets the raw glm object (not avg_slopes) to avoid the duplicate label error from factor variable contrasts. gof_map as list, not string vector -- glm has no r.squared in broom::glance.
> - Q3c: repeat for own_ot

Model actions:

1. Checked educ_resp values: 14 unique codes (1-4 low, 5-8 medium, 9/11/12/1001/1002 high, 97 -> NA affecting 30 obs).
2. Checked glm glance output: confirmed no r.squared column, available stats are null.deviance, df.null, logLik, AIC, BIC, deviance, df.residual, nobs.
3. Wrote `code/03_homeownership_models.R`:
   - educ_cat factor construction
   - LPM via feols with vcov="hetero"
   - Probit via glm with binomial("probit")
   - avg_slopes() for AMEs, saved to CSV
   - modelsummary() with list-based gof_map (nobs + AIC)
   - Repeated for own_ot
4. Ran verification:
   - LPM own: age peak ~66, high educ +5.6pp, unemployment -19.1pp
   - LPM own_ot: steeper education gradient (+21.1pp high), self-emp +18.2pp
   - Noted warnings: non-integer successes (expected), convergence issue on own probit
5. Confirmed 4 output files:
   - `output/tables/q3b_ame_own.csv`
   - `output/tables/q3b_lpm_vs_probit_own.tex`
   - `output/tables/q3c_ame_own_ot.csv`
   - `output/tables/q3c_lpm_vs_probit_own_ot.tex`

Files created: `code/03_homeownership_models.R`, 2 CSVs, 2 .tex files

Prompt 6

User:

> Write 04_merge_proxy.R. Decisions made:
>
> - left_join on educ_resp, print unmatched codes
> - Impute missing p_knows_second_lang with weighted mean of matched values (facine3 weights). Do not use adjacent-code interpolation -- code 97 is unclassifiable.
> - proxy = p_knows_second_lang * as.integer(educ_resp %in% c(9,11,12,1001,1002))
> - Wealth quintiles: use manual wp_fn() cumulative weight approach -- do not use wtd.quantile() from Hmisc (not loaded)
> - Bar chart of mean proxy by wealth quintile, save as PDF
> - Q4c: feols(neg ~ intl_proxy + age_resp + I(age_resp^2) + emp_resp + self_resp + une_resp + hhsize, weights=~facine3, vcov="hetero") For LPM, AME of intl_proxy = its coefficient (linear model). Report mean predicted probability by wealth quintile for context.

Model actions:

1. Inspected secondlang_prob.dta: 12 rows, 2 columns (educ_resp, p_knows_second_lang). Education codes 5 and 7 missing from lookup.
2. Checked neg variable: binary 0/1, label "Owns business (indicator)".
3. Wrote `code/04_merge_proxy.R`:

- left_join, identified unmatched codes 5 and 7
- Imputation with weighted mean (0.376)
- intl_proxy = p_knows_second_lang * high_educ
- Manual weighted quintile function
- Bar chart of proxy by wealth quintile
- LPM of neg on intl_proxy with controls
- Predicted probabilities by wealth quintile

4. Updated `00_master.R`: changed source from "04_robustness.R" to "04_merge_proxy.R"
5. Ran verification:
   - Proxy: Q1=0.073 to Q5=0.291
   - intl_proxy coefficient: 0.028 (p=0.113, not significant)
   - Self-employment dominates at +69.5pp
   - VCOV warning (not positive semi-definite, auto-corrected)
6. Confirmed outputs:
   - `output/figures/q4b_proxy_by_wealth.pdf`
   - `output/tables/q4c_predicted_neg_by_wealth.csv`

Files created: `code/04_merge_proxy.R`, 1 PDF, 1 CSV **Files modified:** `code/00_master.R`

Prompt 7

User:

> Write 05_constructed_variables.R. Variable definitions are fixed:
>
> - fin_share = ifelse(totnet>0, finet/totnet, NA)
> - debt_ratio = ifelse(totnet>0, deud/totnet, NA)
> - non_emp_rate: emp_cols = c("emp_resp", paste0("emp_",2:9)), self_cols = c("self_resp", paste0("self_",2:9)). n_employed = rowSums(==1, na.rm=TRUE) for both, then 1 - n_employed/hhsize, clamped to [0,1]
> - Three groups: owner_one, owner_many, renter as defined above
> - Weighted median via manual cumulative weight function (no Hmisc)
> - Weighted p75 same approach
> - Q5b: stats_fn per group returning mean, median, p75 for both debt_ratio and fin_share
> - Q5c: manual weighted correlation + cor.test() for p-value
> - Q5d: debt_ratio > weighted p75 AND non_emp_rate > 0.5

Model actions:

1. Checked variable distributions: finet (min -3.7M, max 398M), deud (min 0, max 6.8M), totnet > 0 for 6,072 of 6,413 obs. Confirmed all emp/self columns present. hhsize range: 1-10.
2. Wrote `code/05_constructed_variables.R`:
   - fin_share, debt_ratio conditional on totnet > 0
   - non_emp_rate from rowSums, clamped [0,1]
   - Three ownership groups as factor
   - Reusable wq() function for weighted quantiles
   - stats_fn() for group-level statistics
   - Manual weighted correlation for debt_ratio vs fin_share

    - Vulnerability flag: debt_ratio > p75 AND non_emp_rate > 0.5
3. Updated `00_master.R`: changed source from "05_figures.R" to "05_constructed_variables.R"
4. Ran verification:
    - Renters: median fin_share=1.0, median debt=0
    - Owner_one most leveraged (p75 debt=0.551)
    - Weighted correlation (debt vs fin_share): 0.069 (p < 0.001)
    - Vulnerability: owner_one 11.2%, renter 3.9%, owner_many 6.2%
5. Confirmed 3 CSVs saved.

Files created: `code/05_constructed_variables.R`, 3 CSVs **Files modified:** `code/00_master.R`

Prompt 8

User:

> Write 06_mortgage_regressions.R.
>
> - mortgage_holder = as.integer((own==1|own_ot==1) & !is.na(deudre) & deudre>0)
> - Sample: own==1. Rebuild non_emp_rate, fin_share, educ_cat, wealth_q within this script (not relying on Q5 objects).
> - m1: feols(mortgage_holder ~ non_emp_rate + fin_share + age_resp + I(age_resp^2) + educ_cat + hhsize, weights=~facine3, vcov="hetero")
> - m2: add wealth_q
> - m3: add non_emp_rate:fin_share
> - modelsummary() to .tex with coef_map giving clean labels, gof_map as list (N and R-squared), note about weights and robust SEs

Model actions:

1. Checked deudre: no NAs, 1,893 of 5,268 owners have deudre > 0 (35.9%).
2. Wrote `code/06_mortgage_regressions.R`:
    - Self-contained variable construction (not relying on Q5 objects)
    - Three feols models with incremental specifications
    - coef_map with clean labels
    - List-based gof_map (nobs, r.squared)
    - Footnote about weights and robust SEs
3. Updated `00_master.R`: changed source from "06_tables.R" to "06_mortgage_regressions.R"
4. Ran verification:
    - m1: non_emp_rate -19.2pp, fin_share +1.8pp (n.s.)
    - m2: non_emp_rate -20.1pp, fin_share +2.7pp (significant with wealth controls)
    - m3: interaction -0.005 (p=0.84, negligible)
    - Wealth quintiles negative and monotonic
    - 104 obs dropped (totnet <= 0)
5. Confirmed output: `output/tables/q6_mortgage_regressions.tex`

Files created: `code/06_mortgage_regressions.R`, 1 .tex file **Files modified:** `code/00_master.R`

Prompt 9

User:

> Give me a 5-sentence paragraph template for each of the 6 questions. Use [VALUE] for every number. Focus on: what was estimated, what the sign means economically, whether it aligns with theory, and one surprising or noteworthy finding. Do not invent results.

Model actions:

- Wrote six paragraph templates directly, using [VALUE] placeholders for all numerical results.
- Each paragraph covered: estimation method, economic sign interpretation, theoretical alignment, and one noteworthy finding.
- No tools used (pure text generation from session context).

Output: Six paragraphs returned in chat (no files created).

Prompt 9

User:

> Create a log including interaction with the model you have used when solving this task. with prompts history

Model actions:

- Wrote this file (`log/interaction_log_full.md`) with complete prompt history and detailed model actions for each turn.

Summary of all files created

Code (7 files)

File	Purpose
`code/00_master.R`	Master script: loads packages, sets paths, reads data, sources 01-06
`code/01_imputations_weights.R`	Q1: Weight adjustment, Rubin's rules
`code/02_descriptive_stats.R`	Q2: Homeownership by age/gender, bootstrap median, plotting function
`code/03_homeownership_models.R`	Q3: LPM and probit for own and own_ot
`code/04_merge_proxy.R`	Q4: Merge language data, proxy construction, business ownership LPM
`code/05_constructed_variables.R`	Q5: Financial share, debt ratio, non-emp rate, vulnerability
`code/06_mortgage_regressions.R`	Q6: Mortgage holder regressions (3 specifications)

Output tables (11 files)

File	Content
`output/tables/q1b_weight_comparison.csv`	Adjusted vs unadjusted weight comparison
`output/tables/q1d_rubins_rules.csv`	Rubin's combining rules results
`output/tables/q3b_ame_own.csv`	Probit AMEs for main residence
`output/tables/q3b_lpm_vs_probit_own.tex`	LPM vs probit comparison (own)
`output/tables/q3c_ame_own_ot.csv`	Probit AMEs for other real estate
`output/tables/q3c_lpm_vs_probit_own_ot.tex`	LPM vs probit comparison (own_ot)
`output/tables/q4c_predicted_neg_by_wealth.csv`	Predicted business ownership by quintile
`output/tables/q5b_stats_by_owner_group.csv`	Debt and financial share by group
`output/tables/q5c_correlation.csv`	Weighted correlation results
`output/tables/q5d_vulnerability.csv`	Vulnerability rates by group
`output/tables/q6_mortgage_regressions.tex`	Three-model LaTeX regression table

Output figures (4 files)

File	Content
`output/figures/q2b_homeownership_age_gender.pdf`	Homeownership by age and gender (3 series)
`output/figures/q2c_median_wealth_bootstrap.pdf`	Bootstrap median wealth with CI ribbon
`output/figures/q2d_other_realestate_ownership.pdf`	Other real estate ownership (function demo)
`output/figures/q4b_proxy_by_wealth.pdf`	International proxy by wealth quintile

Report and logs (2 files)

File	Content
`report.Rmd`	RMarkdown source for full analysis report
`report.pdf`	Compiled PDF report (103 KB)
`log/interaction_log_full.md`	This file: full prompt history and model actions

Technical notes

1. **R version**: 4.5 (ARM64 Mac)

2. **LaTeX**: TinyTeX at `/Users/sid/Library/TinyTeX`
3. **Pandoc**: `/usr/local/bin/pandoc`
4. **Key packages**: haven 2.5+, tidyverse 2.0, fixest 0.12+, modelsummary 2.0+, boot 1.3, marginaleffects 0.25+, kableExtra 1.4
5. **Warnings handled**:
   - Non-integer successes in binomial glm (expected with survey weights)
   - Probit convergence failure for `own` (quasi-separation)
   - VCOV not positive semi-definite in Q4 (auto-corrected by fixest)
   - marginaleffects AME SEs return NA with non-integer weights
6. **All scripts use implicate 1** except 01_imputations_weights.R which uses all 5. For final publication, Rubin's rules should be applied across all implicates for each estimand.